

Team: Sakka Mohamad-Mario, Mirghani Abdelrahman 1241EB

Exercises 2

Active ontology × Entities × Individuals by class × DL Query ×

Ontology header: ? ⌵ ⌶ ⌷ ⌸

Ontology IRI

Ontology Version IRI

Annotations +

rdfs:comment
A pizza ontology that describes various pizzas based on their toppings. × ⌵

Exercise 3

Active ontology × Entities × Object properties × Individuals by class × DL Query ×

Ontology header: ? ⌵ ⌶ ⌷ ⌸

Ontology IRI

Ontology Version IRI

Annotations +

rdfs:label
Pizza ontology exercise snapshot × ⌵

rdfs:comment
A pizza ontology that describes various pizzas based on their toppings. × ⌵

Ontology metrics: ? ⌵ ⌶ ⌷ ⌸

Metrics

Axiom	0
Logical axiom count	0
Declaration axioms count	0
Class count	0
Object property count	0
Data property count	0
Individual count	0
Annotation Property count	2

Class axioms

SubClassOf	0
EquivalentClasses	0
DisjointClasses	0
GCI count	0
Hidden GCI Count	0

Object property axioms

Ontology imports ? ⌵ ⌶ ⌷ ⌸

Ontology Prefixes General class axioms

Imported ontologies: ? ⌵ ⌶ ⌷ ⌸

Direct Imports +

Indirect Imports

Exercise 4

Class hierarchy (inferred)

Class hierarchy

Class hierarchy: :Pizza

owl:Thing

:Pizza

:PizzaBase

:PizzaTopping

Asserted

Annotations

Usage

Annotations: :PizzaTopping

Annotations

Description: :PizzaTopping

Equivalent To

SubClass Of

owl:Thing

Exercise 5

Active ontology × Entities × Individuals by class × DL Query ×

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: :Pizza     

    Asserted ▼

owl:Thing
 :Pizza
 :PizzaBase
 :PizzaTopping

Annotations Usage

Annotations: :Pizza

Annotations 

Description: :Pizza

SubClass Of 
  owl:Thing

General class axioms 

SubClass Of (Anonymous Ancestor)

Instances 

Target for Key 

Disjoint With 
  :PizzaBase,  :PizzaTopping

Exercise 6

Active ontology × Entities × **Classes** × Object properties × Data properties × Annotation pro

Class hierarchy Class hierarchy (inferred) Annotations Usage

Class hierarchy: :DeepPanBase Annotations: :DeepPanBase

Asserted

- owl:Thing
 - :Pizza
 - :PizzaBase
 - :DeepPanBase**
 - :ThinAndCrispyBase
 - :PizzaTopping

Annotations +

Description: :DeepPanBase

Equivalent To +

SubClass Of +

- :PizzaBase

General class axioms +

SubClass Of (Anonymous Ancestor)

Instances +

Target for Key +

Disjoint With +

- :ThinAndCrispyBase

Exercise 7

Active ontology x Entities x Individuals by class x DL Query x

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: :JalapenoPepperTopping ? [] [] [] [] []

Asserted

- owl:Thing
 - :Pizza
 - :PizzaBase
 - :PizzaTopping
 - :CheeseTopping
 - :MozzarellaTopping
 - :ParmesanTopping
 - :MeatTopping
 - :PepperoniTopping
 - :SpicyBeefTopping
 - :SeafoodTopping
 - :AnchovyTopping
 - :PrawnTopping
 - :TunaTopping
 - :VegetableTopping
 - :CaperTopping
 - :GarlicTopping
 - :MushroomTopping
 - :OliveTopping
 - :OnionTopping
 - :PepperTopping
 - :JalapenoPepperTopping**
 - :RedPepperTopping
 - :GreenPepperTopping
 - :TomatoTopping

Annotations Usage

Annotations: :JalapenoPepperTopping

Annotations +

Description: :JalapenoPepperTopping

Equivalent To +

SubClass Of +

:PepperTopping

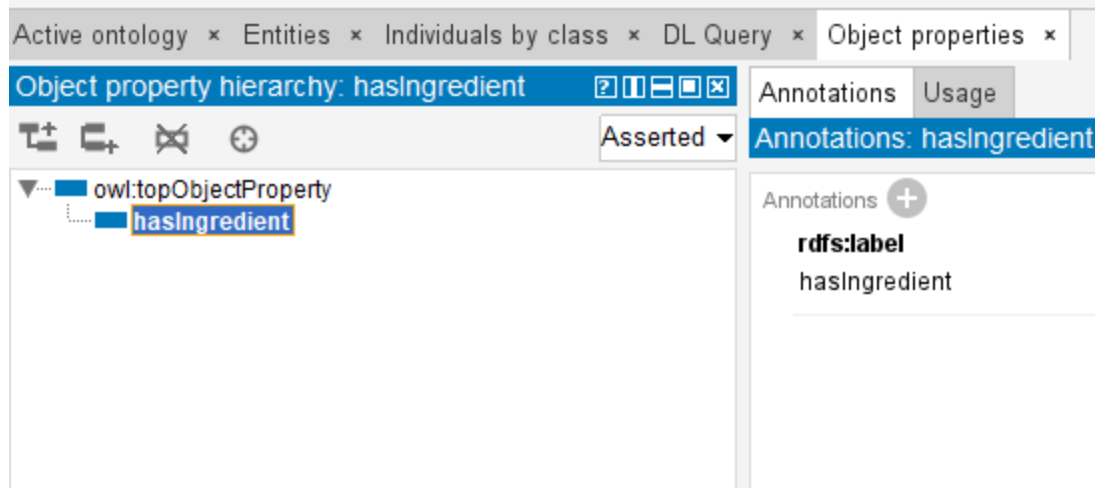
General class axioms +

SubClass Of (Anonymous Ancestor)

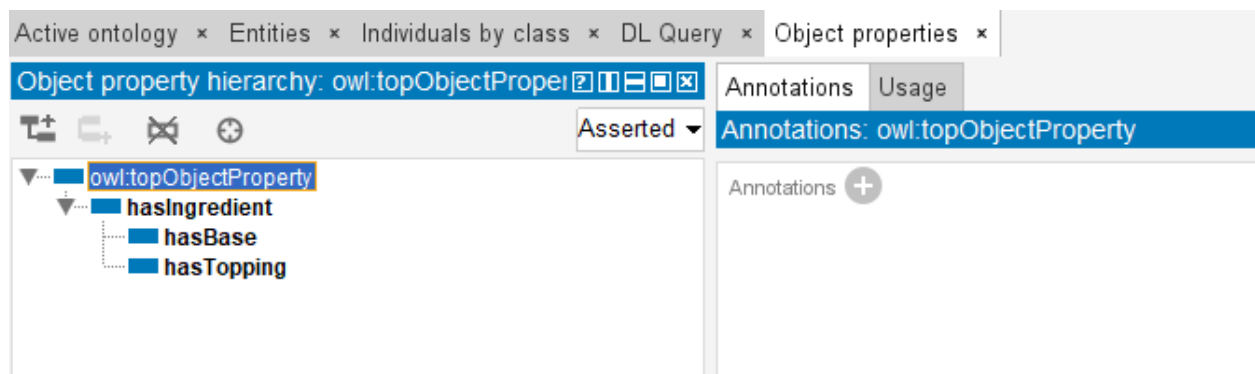
Instances +

Target for Key +

Exercise 8



Exercise 9



Exercise 10

Object property hierarchy: isBaseOf

Annotations Usage

Annotations: isBaseOf

Annotations +

rdfs:label
isBaseOf

owl:topObjectProperty

- hasIngredient
 - hasBase
 - hasTopping
- isBaseOf**
- isIngredientOf
- isToppingOf

Chara ? ? ? ? ?

Description: isBaseOf

☐ Functional

☐ Inverse functiona

☐ Transitive

☐ Symmetric

☐ Asymmetric

Equivalent To +

SubProperty Of +

Inverse Of +

hasBase

Exercise 11

Active ontology × Entities × Object properties × Individuals by class × DL Query ×

Object property hierarchy: :hasIngredient

Annotations Usage

Annotations: :hasIngredient

Annotations +

Characteristics: :hasIngredient

☐ Functional
☐ Inverse functional
☒ Transitive
☐ Symmetric
☐ Asymmetric

Equivalent To +
 SubProperty Of +
 Inverse Of +
 isIngredientOf

isIngredientOf

Active ontology × Entities × Object properties × Individuals by class × DL Query ×

Object property hierarchy: isIngredientOf

Annotations Usage

Annotations: isIngredientOf

Annotations +

rdfs:label
 isIngredientOf

Characteristics: isIngredientOf

☐ Functional
☐ Inverse functional
☒ Transitive
☐ Symmetric
☐ Asymmetric

Equivalent To +
 SubProperty Of +
 Inverse Of +
 :hasIngredient

Exercise 12

Active ontology × Entities × Object properties × Individuals by class × DL Query ×

Object property hierarchy: hasBase [?] [I] [E] [Q] [X]

Annotations Usage

Annotations: hasBase

Annotations +

rdfs:label
hasBase

owl:topObjectProperty

- hasIngredient
 - hasBase
 - hasTopping
- isBaseOf
- isIngredientOf
- isToppingOf

Charac [?] [I] [E] [Q] [X] Description: hasBase

☒ Functional Equivalent To +

Exercise 13

Active ontology × Entities × Object properties × Individuals by class × DL Query ×

Object property hierarchy: hasTopping

Annotations Usage

Annotations: hasTopping

Annotations +

rdfs:label
hasTopping

owl:topObjectProperty

- hasIngredient
 - hasBase
 - hasTopping**
 - isBaseOf
 - isIngredientOf
 - isToppingOf

Chara Description: hasTopping

☐ Functional
☐ Inverse functiona
☐ Transitive
☐ Symmetric
☐ Asymmetric
☐ Reflexive
☐ Irreflexive

Equivalent To +

SubProperty Of +

hasIngredient

Inverse Of +

isToppingOf

Domains (intersection) +

Ranges (intersection) +

PizzaTopping

Exercise 14

Active ontology × Entities × Object properties × Individuals by class × DL Query ×

Object property hierarchy: hasTopping

Annotations Usage

Annotations: hasTopping

Annotations +

rdfs:label
hasTopping

Charac? || || || || || Description: hasTopping

☐ Functional

☐ Inverse functional

☐ Transitive

☐ Symmetric

☐ Asymmetric

☐ Reflexive

☐ Irreflexive

Equivalent To +

SubProperty Of +

:hasIngredient

Inverse Of +

:isToppingOf

Domains (intersection) +

:Pizza

Ranges (intersection) +

:PizzaTopping

owl:topObjectProperty

:hasIngredient

hasBase

hasTopping

isBaseOf

isIngredientOf

isToppingOf

Asserted

Exercise 15

Active ontology × Entities × Object properties × Individuals by class × DL Query ×

Object property hierarchy: hasBase

Annotations Usage

Annotations: hasBase

Annotations +

rdfs:label
hasBase

Chara

Description: hasBase

☒ Functional

☐ Inverse functiona

☐ Transitive

☐ Symmetric

☐ Asymmetric

☐ Reflexive

☐ Irreflexive

Equivalent To +

SubProperty Of +

hasIngredient

Inverse Of +

isBaseOf

Domains (intersection) +

Pizza

Ranges (intersection) +

PizzaBase

- owl:topObjectProperty
 - :hasIngredient
 - hasBase
 - hasTopping
 - isBaseOf
 - isIngredientOf
 - isToppingOf

Annotations +

rdfs:label
 isBaseOf

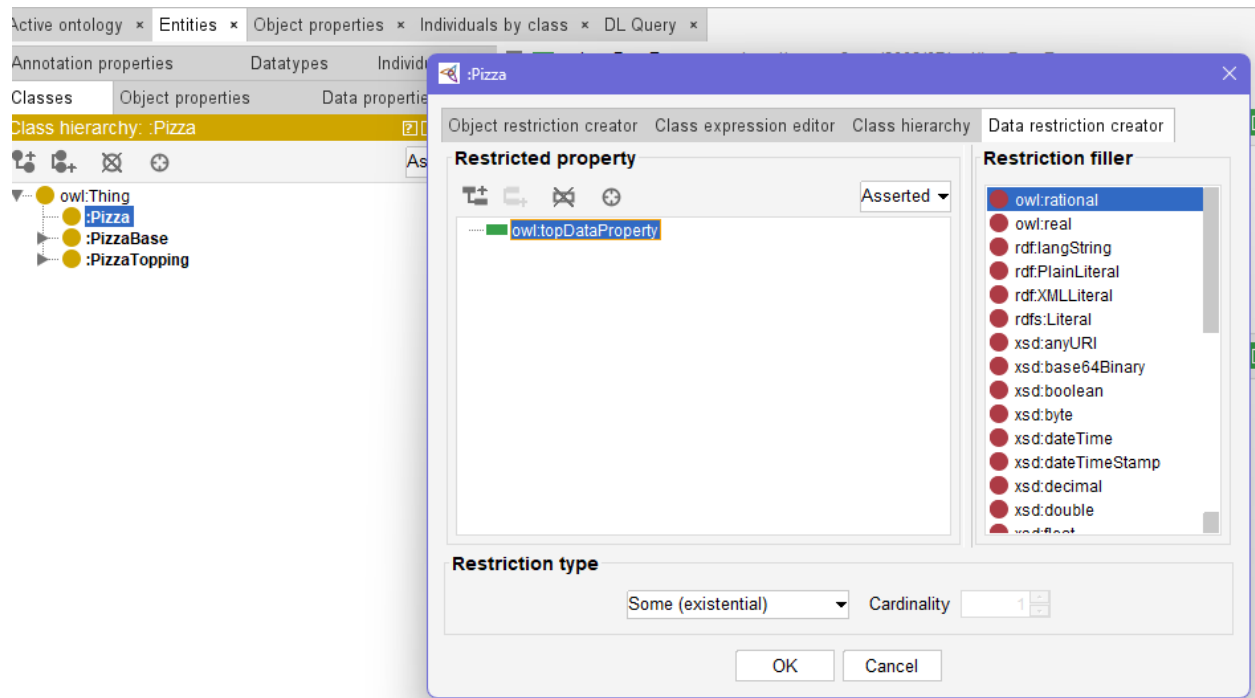
Chara

☐ Functional
☐ Inverse functiona
☐ Transitive
☐ Symmetric
☐ Asymmetric
☐ Reflexive
☐ Irreflexive

Description: isBaseOf

Equivalent To +
 SubProperty Of +
 Inverse Of +
 hasBase
 Domains (intersection) +
 :PizzaBase
 Ranges (intersection) +
 :Pizza

Exercise 16



Exercise 17

Annotation properties

Datatypes

Individuals

Classes

Object properties

Data properties

Class hierarchy: :Pizza

owl:Thing

:Pizza

:PizzaBase

:PizzaTopping

Asserted

Annotations

Usage

Annotations: :Pizza

Annotations

Description: :Pizza

Equivalent To

SubClass Of

hasBase some :PizzaBase

owl:Thing

General class axioms

SubClass Of (Anonymous Ancestor)

Instances

Target for Key

Disjoint With

:PizzaBase, :PizzaTopping

Exercise 18

Active ontology x Entities x Object properties x Individuals by class x DL Query x

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: :MargheritaPizza

Annotations: :MargheritaPizza

Annotations

rdfs:comment
A pizza that only has Mozzarella and Tomato toppings.

Description: :MargheritaPizza

Equivalent To

SubClass Of

:NamedPizza

General class axioms

SubClass Of (Anonymous Ancestor)

hasBase some :PizzaBase

Class hierarchy: :MargheritaPizza

owl:Thing
:Pizza
:NamedPizza
:MargheritaPizza
:PizzaBase
:PizzaTopping

Exercise 19

Active ontology × Entities × Object properties × Individuals by class × DL Query ×

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: :MargheritaPizza

- owl:Thing
 - :Pizza
 - :NamedPizza
 - :MargheritaPizza**
 - :PizzaBase
 - :PizzaTopping
 - :CheeseTopping
 - :MozzarellaTopping
 - :ParmesanTopping
 - :MeatTopping
 - :SeafoodTopping
 - :VegetableTopping

Annotations Usage

Annotations: :MargheritaPizza

Annotations +

rdfs:comment
A pizza that only has Mozzarella and Tomato toppings.

Description: :MargheritaPizza

Equivalent To +

SubClass Of +

- :NamedPizza
- hasTopping some :MozzarellaTopping

General class axioms +

SubClass Of (Anonymous Ancestor)

- hasBase some :PizzaBase

Exercise 20

Active ontology x Entities x Object properties x Individuals by class x DL Query x

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: :MargheritaPizza

owl:Thing

- :Pizza
 - :NamedPizza
 - :MargheritaPizza
 - :PizzaBase
 - :DeepPanBase
 - :ThinAndCrispyBase
 - :PizzaTopping
 - :CheeseTopping
 - :MeatTopping
 - :SeafoodTopping
 - :VegetableTopping

Asserted

:MargheritaPizza — http://www.pizza.com/ontologies/pizza.owl#MargheritaPizza

Annotations Usage

Annotations: :MargheritaPizza

Annotations +

rdfs:comment

A pizza that only has Mozzarella and Tomato toppings.

Description: :MargheritaPizza

Equivalent To +

SubClass Of +

- :NamedPizza
- hasTopping some :MozzarellaTopping
- hasTopping some :TomatoTopping

General class axioms +

SubClass Of (Anonymous Ancestor)

- hasBase some :PizzaBase

Instances +

Exercise 21

Active ontology x Entities x Object properties x Individuals by class x DL Query x

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: :AmericanaPizza

owl:Thing

- :Pizza
 - :NamedPizza
 - :AmericanaPizza
 - :MargheritaPizza
 - :PizzaBase
 - :PizzaTopping

Asserted

:AmericanaPizza — http://www.pizza.com/ontologies/pizza.owl#AmericanaPizza

Annotations Usage

Annotations: :AmericanaPizza

Annotations +

Description: :AmericanaPizza

Equivalent To +

SubClass Of +

- :NamedPizza
- hasTopping some :MozzarellaTopping
- hasTopping some :PepperoniTopping
- hasTopping some :TomatoTopping

General class axioms +

SubClass Of (Anonymous Ancestor)

- hasBase some :PizzaBase

Exercise 22

Active ontology × Entities × Object properties × Individuals by class × DL Query ×

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: :SohoPizza

owl:Thing

- :Pizza
 - :NamedPizza
 - :AmericanaPizza
 - :AmericanHotPizza
 - :MargheritaPizza
 - :SohoPizza**
 - :PizzaBase
 - :DeepPanBase
 - :ThinAndCrispyBase
 - :PizzaTopping
 - :CheeseTopping
 - :MeatTopping
 - :SeafoodTopping
 - :VegetableTopping

Annotations: :SohoPizza

Annotations +

Description: :SohoPizza

Equivalent To +

SubClass Of +

- :NamedPizza
- hasTopping some :MozzarellaTopping
- hasTopping some :OliveTopping
- hasTopping some :ParmesanTopping
- hasTopping some :TomatoTopping

General class axioms +

SubClass Of (Anonymous Ancestor)

- hasBase some :PizzaBase

Exercise 23

Active ontology × Entities × Object properties × Individuals by class × DL Query ×

Annotation propertiesDatatypesIndividuals

ClassesObject propertiesData properties

Class hierarchy: :AmericanaPizza

owl:Thing

:Pizza

:NamedPizza

:AmericanaPizza

:AmericanHotPizza

:MargheritaPizza

:SohoPizza

:PizzaBase

:PizzaTopping

AnnotationsUsage

Annotations: :AmericanaPizza

Annotations

Description: :AmericanaPizza

Equivalent To

SubClass Of

:NamedPizza

hasTopping some :MozzarellaTopping

hasTopping some :PepperoniTopping

hasTopping some :TomatoTopping

General class axioms

SubClass Of (Anonymous Ancestor)

hasBase some :PizzaBase

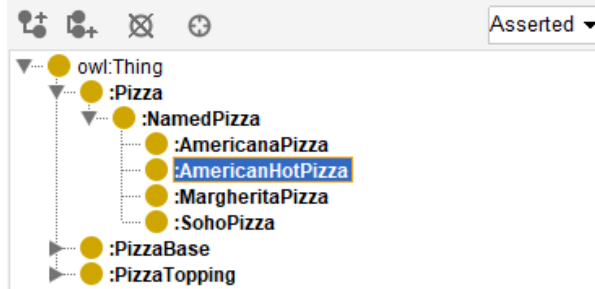
Instances

Target for Key

Disjoint With

:MargheritaPizza, :SohoPizza, :AmericanHotPizza

Class hierarchy: :AmericanHotPizza



Annotations: :AmericanHotPizza

Annotations +

Description: :AmericanHotPizza

Equivalent To +

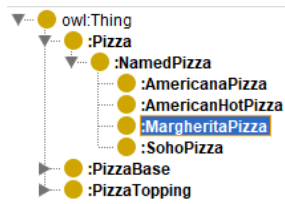
SubClass Of +

- :NamedPizza
- hasTopping some :JalapenoPepperTopping
- hasTopping some :MozzarellaTopping
- hasTopping some :PepperoniTopping
- hasTopping some :TomatoTopping

General class axioms +

SubClass Of (Anonymous Ancestor)

- hasBase some :PizzaBase



rdfs:comment

A pizza that only has Mozzarella and Tomato toppings.

Description: :MargheritaPizza

Equivalent To +

SubClass Of +

● :NamedPizza

● hasTopping some :MozzarellaTopping

● hasTopping some :TomatoTopping

General class axioms +

SubClass Of (Anonymous Ancestor)

● hasBase some :PizzaBase

Instances +

Target for Key +

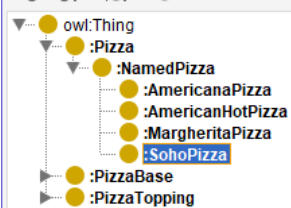
Disjoint With +

● :AmericanaPizza, :SohoPizza, :AmericanHotPizza

Classes Object properties Data properties

Class hierarchy: :SohoPizza

Class hierarchy icons: owl, pizza, named pizza, americana pizza, american hot pizza, margherita pizza, soho pizza, pizza base, pizza topping. Asserted



Annotations Usage

Annotations: :SohoPizza

Annotations +

Description: :SohoPizza

Equivalent To +

SubClass Of +

● :NamedPizza

● hasTopping some :MozzarellaTopping

● hasTopping some :OliveTopping

● hasTopping some :ParmesanTopping

● hasTopping some :TomatoTopping

? @
? @
? @
? @
? @

Exercise 24

Classes

Object properties

Data properties

Class hierarchy: :ProbelInconsistentTopping

owl:Thing

:Pizza

:PizzaBase

:PizzaTopping

:CheeseTopping

:MozzarellaTopping

:ParmesanTopping

:ProbelInconsistentTopping

:MeatTopping

:SeafoodTopping

:VegetableTopping

Annotations

Usage

Annotations: :ProbelInconsistentTopping

Annotations

rdfs:comment

This class should be inconsistent when the ontology is cla

Description: :ProbelInconsistentTopping

Equivalent To

SubClass Of

General class axioms

:CheeseTopping

:VegetableTopping

Exercise 25

Classes

Object properties

Data properties

Class hierarchy: :ProbelInconsistentTopping

owl:Thing

:Pizza

:PizzaBase

:PizzaTopping

:CheeseTopping

:MozzarellaTopping

:ParmesanTopping

:ProbelInconsistentTopping

:MeatTopping

:SeafoodTopping

:VegetableTopping

Annotations

Usage

Annotations: :ProbelInconsistentTopping

Annotations

rdfs:comment

This class should be inconsistent when the ont

Description: :ProbelInconsistentTopping

Equivalent To

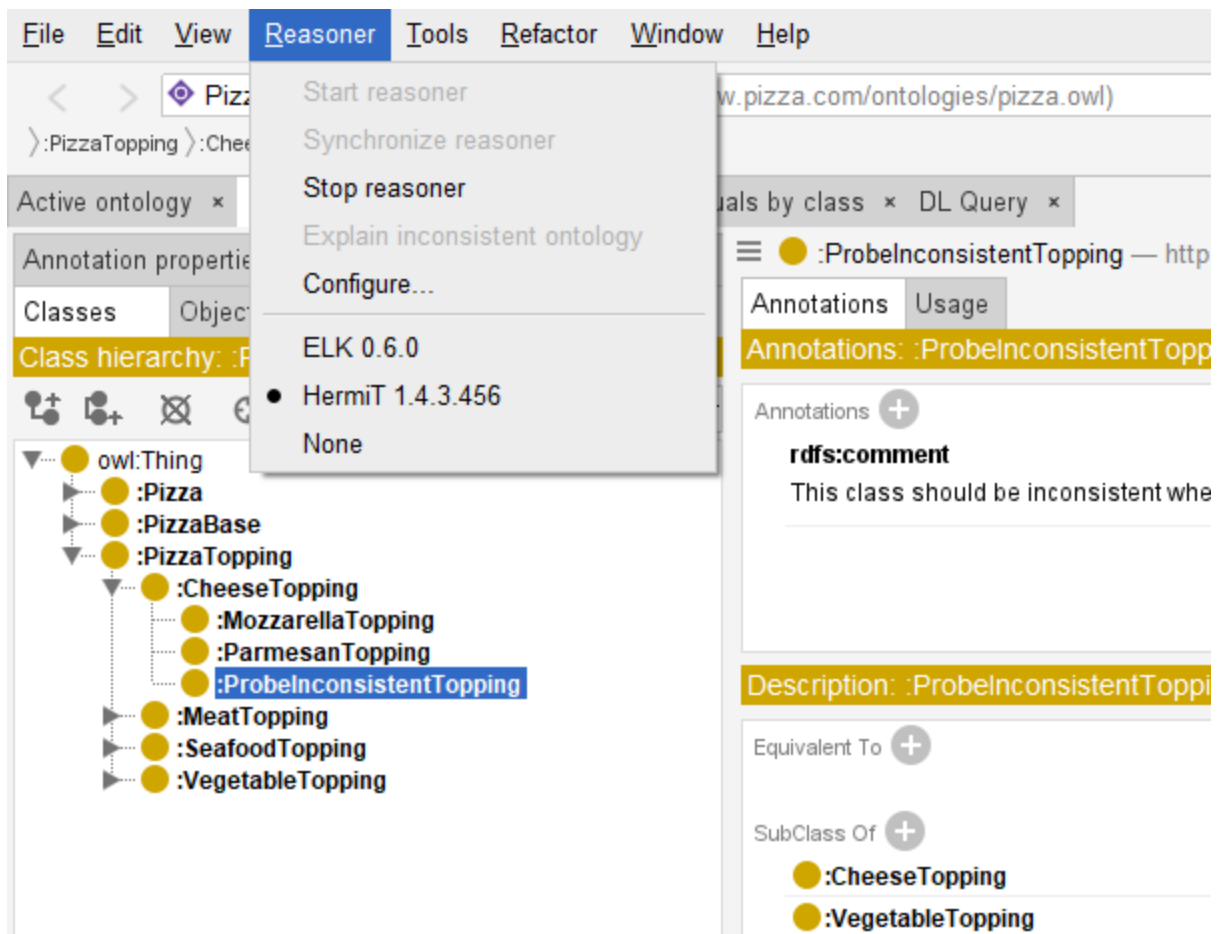
owl:Nothing

SubClass Of

:CheeseTopping

:VegetableTopping

Exercise 26



Exercise 27

Classes

Object properties

Data properties

Class hierarchy: :ProbelnconsistentTopping

owl:Thing

:Pizza

:PizzaBase

:PizzaTopping

:CheeseTopping

:MozzarellaTopping

:ParmesanTopping

:ProbelnconsistentTopping

:MeatTopping

:SeafoodTopping

:VegetableTopping

Annotations

Usage

Annotations: :ProbelnconsistentTopping

Annotations

rdfs:comment

This class should be inconsistent when the ontology

Description: :ProbelnconsistentTopping

Equivalent To

owl:Nothing

SubClass Of

:CheeseTopping

:VegetableTopping

:ParmesanTopping

:ProbelnconsistentTopping

Exercise 28

Classes	Object properties	Data properties	Annotations	Usage
Class hierarchy: :CheesyPizza			Annotations: :CheesyPizza	
<pre> graph TD owlThing[owl:Thing] --> Pizza[Pizza] Pizza --> CheesyPizza[CheesyPizza] Pizza --> NamedPizza[NamedPizza] Pizza -- PizzaBase[PizzaBase] Pizza -- PizzaTopping[PizzaTopping] </pre>			Annotations + Description: :CheesyPizza Equivalent To + SubClass Of + • :Pizza • hasTopping some :CheeseTopping General class axioms + SubClass Of (Anonymous Ancestor) • hasBase some :PizzaBase	

Exercise 29

Classes	Object properties	Data properties	Annotations	Usage
Class hierarchy: :CheesyPizza			Annotations: :CheesyPizza	
<pre> graph TD owlThing[owl:Thing] --> Pizza[Pizza] Pizza --> CheesyPizza[CheesyPizza] Pizza --> NamedPizza[NamedPizza] NamedPizza --> AmericanaPizza[AmericanaPizza] NamedPizza --> AmericanHotPizza[AmericanHotPizza] NamedPizza --> MargheritaPizza[MargheritaPizza] NamedPizza --> SohoPizza[SohoPizza] Pizza -- PizzaBase[PizzaBase] Pizza -- PizzaTopping[PizzaTopping] </pre>			Annotations + Description: :CheesyPizza Equivalent To + • :Pizza and (hasTopping some :CheeseTopping) SubClass Of + • :Pizza General class axioms +	

Exercise 30

File Edit View Reasoner Tools Refactor Window Help

Pizza ontology exercise snapshot (http://www.pizza.com/ontologies/pizza.owl)

>:Pizza >:CheesyPizza

Active ontology * Entities * Object properties * Individuals by class * DL Query *

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: :CheesyPizza

Annotations: :CheesyPizza

Annotations +

Description: :CheesyPizza

Equivalent To +

:Pizza and (hasTopping some :CheeseTopping)

SubClass Of +

:Pizza

owl:Thing

:Pizza

:CheesyPizza

:NamedPizza

:AmericanaPizza

:AmericanHotPizza

:MargheritaPizza

:SohoPizza

:PizzaBase

:PizzaTopping

Exercise 31

Classes Object properties Data properties

Class hierarchy: :VegetarianPizza

Annotations: :VegetarianPizza

Annotations +

Description: :VegetarianPizza

Equivalent To +

:Pizza and (hasTopping only (:CheeseTopping or :VegetableTopping))

SubClass Of +

General class axioms +

SubClass Of (Anonymous Ancestor)

hasBase some :PizzaBase

owl:Thing

:Pizza

:CheesyPizza

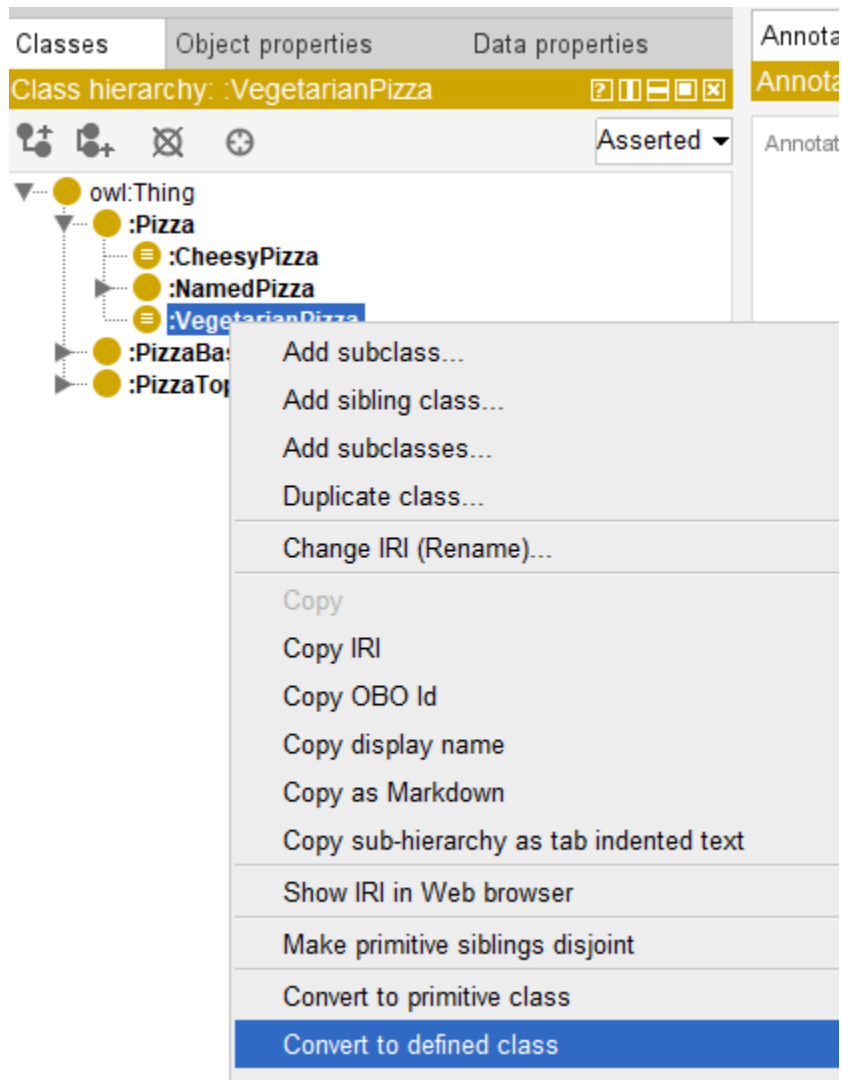
:NamedPizza

:VegetarianPizza

:PizzaBase

:PizzaTopping

Exercise 32



Exercise 33

Classes	Object properties	Data properties	Annotations	Usage
Class hierarchy: :VegetarianPizza			Annotations: :VegetarianPizza	
<pre> graph TD owlThing[owl:Thing] --> Pizza[Pizza] Pizza --> CheesyPizza[CheesyPizza] CheesyPizza --> NamedPizza[NamedPizza] NamedPizza --> VegetarianPizza[VegetarianPizza] PizzaBase[PizzaBase] --> Pizza PizzaTopping[PizzaTopping] --> Pizza </pre>			Annotations +	
			Description: :VegetarianPizza Equivalent To + :Pizza and (hasTopping only (:CheeseTopping or :VegetableTopping)) SubClass Of + :Pizza General class axioms + SubClass Of (Anonymous Ancestor) hasBase some :PizzaBase	

Exercise 34

Classes

Object properties

Data properties

Annotations

Usage

Class hierarchy: :MargheritaPizza

owl:Thing

:Pizza

:CheesyPizza

:NamedPizza

:AmericanaPizza

:AmericanHotPizza

:MargheritaPizza

:SohoPizza

:VegetarianPizza

:PizzaBase

:PizzaTopping

Annotations: :MargheritaPizza

Annotations

rdfs:comment

A pizza that only has Mozzarella and Tomato toppings.

Description: :MargheritaPizza

Equivalent To

SubClass Of

:NamedPizza

hasTopping **only**

(:MozzarellaTopping **or** :TomatoTopping)

hasTopping **some** :MozzarellaTopping

hasTopping **some** :TomatoTopping

General class axioms

SubClass Of (Anonymous Ancestor)

hasBase **some** :PizzaBase

Exercise 35



Exercise 36

Classes

Object properties

Data properties

Class hierarchy: :AmericanaPizza

?

⌵

⌵

⌵

⌵

Asserted

- owl:Thing
 - :Pizza
 - :CheesyPizza
 - :NamedPizza
 - :AmericanaPizza**
 - :AmericanHotPizza
 - :MargheritaPizza
 - :SohoPizza
 - :VegetarianPizza
 - :PizzaBase
 - :PizzaTopping

Annotations

Usage

Annotations: :AmericanaPizza

Annotations +

Description: :AmericanaPizza

Equivalent To +

SubClass Of +

- :NamedPizza
- hasTopping **only** (:MozzarellaTopping or :PepperoniTopping or :TomatoTopping)
- hasTopping **some** :MozzarellaTopping
- hasTopping **some** :PepperoniTopping
- hasTopping **some** :TomatoTopping

General class axioms +

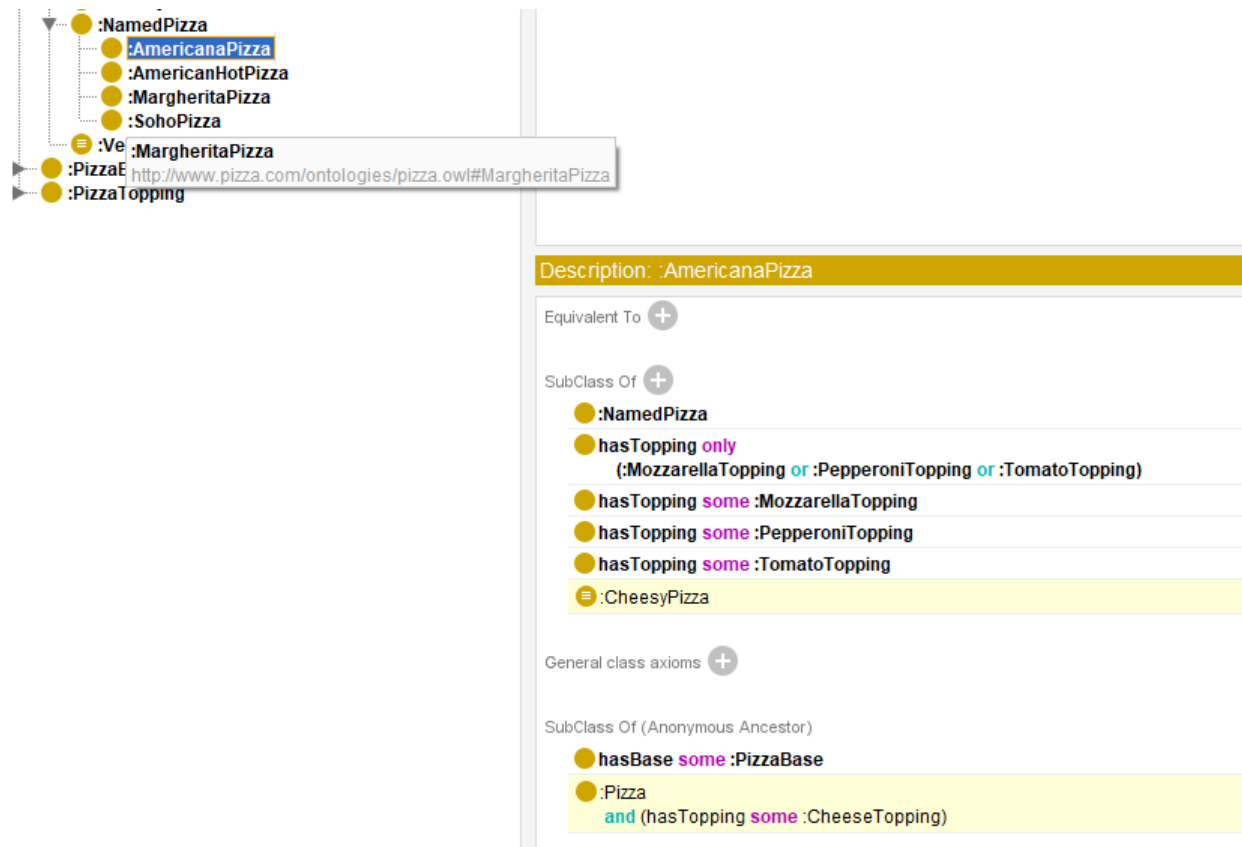
SubClass Of (Anonymous Ancestor)

- hasBase **some** :PizzaBase

Exercise 37



Exercise 38



Exercise 39

Object property hierarchy: hasSpiciness

Annotations

Usage

Annotations: hasSpiciness

Annotations

+

rdfs:label

hasSpiciness

Chara

Description: hasSpiciness

☒ Functional
 ☐ Inverse function
 ☐ Transitive
 ☐ Symmetric
 ☐ Asymmetric
 ☐ Reflexive
 ☐ Irreflexive

Equivalent To

+

SubProperty Of

+

Inverse Of

+

Domains (intersection)

+

Ranges (intersection)

+

⊔ :SpicinessValuePartition

Classes

Object properties

Data properties

Class hierarchy: :SpicinessValuePartition

Annotations

Usage

Annotations: :SpicinessValuePartition

Annotations

+

Description: :SpicinessValuePartition

Equivalent To

+

⊔ :Hot or :Medium or :Mild

SubClass Of

+

⊔ :ValuePartition

owl:Thing

⊔ :Pizza

⊔ :PizzaBase

⊔ :PizzaTopping

⊔ :ValuePartition

⊔ :SpicinessValuePartition

⊔ :Hot

⊔ :Medium

⊔ :Mild

Exercise 40

Description: :JalapenoPepperTopping	
Equivalent To	
SubClass Of	
	 :PepperTopping
	 hasSpiciness some  :Hot
General class axioms	
SubClass Of (Anonymous Ancestor)	
Instances	
Target for Key	
Disjoint With	
	 :SweetPepperTopping

Exercise 41

Classes

Object properties

Data properties

Class hierarchy: :SpicyPizza

owl:Thing

:Pizza

:CheesyPizza

:NamedPizza

:SpicyPizza

:VegetarianPizza

:PizzaBase

:PizzaTopping

:ValuePartition

Annotations

Usage

Annotations: :SpicyPizza

Annotations

Description: :SpicyPizza

Equivalent To

:Pizza

and (hasTopping some (:PizzaTopping and (hasSpiciness some :Hot)))

SubClass Of

General class axioms

SubClass Of (Anonymous Ancestor)

hasBase some :PizzaBase

Exercise 42

Class hierarchy: :SpicyPizza

owl:Thing

:Pizza

:CheesyPizza

:NamedPizza

:SpicyPizza

:VegetarianPizza

:PizzaBase

:PizzaTopping

:ValuePartition

Annotations: :SpicyPizza

Annotations +

Description: :SpicyPizza

Equivalent To +

:Pizza

and (hasTopping some (:PizzaTopping and (hasSpiciness some :Hot)))

SubClass Of +

:Pizza

General class axioms +

SubClass Of (Anonymous Ancestor)

hasBase some :PizzaBase

Exercise 43

Classes

Object properties

Data properties

Class hierarchy: :InterestingPizza

owl:Thing

:Pizza

:CheesyPizza

:InterestingPizza

:NamedPizza

:SpicyPizza

:VegetarianPizza

:PizzaBase

:PizzaTopping

:ValuePartition

Annotations

Usage

Annotations: :InterestingPizza

Annotations +

Description: :InterestingPizza

Equivalent To +
:Pizza
and (hasTopping min 3 owl:Thing)

SubClass Of +

General class axioms +

SubClass Of (Anonymous Ancestor)
hasBase some :PizzaBase

Exercise 44

Classes

Object properties

Data properties

Annotations

Usage

Class hierarchy: :InterestingPizza

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Annotations: :InterestingPizza

Annotations +

Description: :InterestingPizza

Equivalent To +

:Pizza

and (hasTopping min 3 owl:Thing)

SubClass Of +

:Pizza

General class axioms +

SubClass Of (Anonymous Ancestor)

hasBase some :PizzaBase

Exercise 45

Classes	Object properties	Data properties	Annotations	Usage
Class hierarchy: :FourCheesePizza			Annotations: :FourCheesePizza	
<pre> graph TD owlThing[owl:Thing] --> Pizza[:Pizza] Pizza --> CheesyPizza[:CheesyPizza] Pizza --> FourCheesePizza[:FourCheesePizza] Pizza --> InterestingPizza[:InterestingPizza] Pizza --> NamedPizza[:NamedPizza] Pizza --> SpicyPizza[:SpicyPizza] Pizza --> VegetarianPizza[:VegetarianPizza] Pizza --> PizzaBase[:PizzaBase] Pizza --> PizzaTopping[:PizzaTopping] Pizza --> ValuePartition[:ValuePartition] </pre>			Annotations + Description: :FourCheesePizza Equivalent To + SubClass Of + • :Pizza • hasTopping exactly 4 :CheeseTopping General class axioms + SubClass Of (Anonymous Ancestor) • hasBase some :PizzaBase	

Exercise 46

Classes	Object properties	Data properties	Annotations
Data property hierarchy: hasCalorificContent			Annotations
<pre> graph TD owlTopDataProperty[owl:topDataProperty] --> hasCalorificContentValue[hasCalorificContentValue] </pre>			Annotations hasCalorificContentValue

Exercise 47

Classes	Object properties	Data properties	Annotations	Usage
Individuals: :QuattroFormaggio			Annotations: :QuattroFormaggio	
◆ :ExampleAmericana ◆ :ExampleMargherita ◆ :ExampleSoho ◆ :QuattroFormaggio			Annotations +	
Description: :QuattroFormaggio			Property assertions: :QuattroFormaggio	
Types + ● :Pizza			Object property assertions + Data property assertions + hasCalorificContentValue 723	
Same Individual As + Different Individuals +				

Exercise 48

Classes	Object properties	Data properties	Annotations	Usage
Class hierarchy: :Pizza			Annotations: :Pizza	
Asserted			Annotations +	
- owl:Thing :Pizza :CheesyPizza :FourCheesePizza :InterestingPizza :NamedPizza :SpicyPizza :VegetarianPizza :PizzaBase :PizzaTopping :ValuePartition			Description: :Pizza	
			Equivalent To + SubClass Of + - hasBase some :PizzaBase - hasCalorificContentValue some xsd:integer - owl:Thing General class axioms + SubClass Of (Anonymous Ancestor) Instances + ◆ :QuattroFormaggio	

Exercise 49

Classes

Object properties

Data properties

Class hierarchy: :HighCaloriePizza

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Annotations

Usage

Annotations: :HighCaloriePizza

Annotations

+

Description: :HighCaloriePizza

Equivalent To

+

:Pizza

and (hasCalorificContentValue some xsd:integer[>= 400])

SubClass Of

+

Class hierarchy: :LowCaloriePizza

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Description: :HighCaloriePizza

Equivalent To 

 **:Pizza**
and (hasCalorificContentValue  xsd:integer[>= 400])

SubClass Of 

 **:Pizza**

General class axioms 

SubClass Of (Anonymous Ancestor)

 **hasBase**  **:PizzaBase**

 **hasCalorificContentValue**  xsd:integer

Instances 

 **:ExampleAmericana**

 **:QuattroFormaggio**

Description: :LowCaloriePizza

Equivalent To 

 **:Pizza**
and (hasCalorificContentValue  xsd:integer[< 400])

SubClass Of 

 **:Pizza**

General class axioms 

SubClass Of (Anonymous Ancestor)

 **hasBase**  **:PizzaBase**

 **hasCalorificContentValue**  xsd:integer

Instances 

 **:ExampleMargherita**

 **:ExampleSoho**

Exercise 51

Active ontology × Entities × Object properties × Individuals by class × DL Query ×

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: :NonVegetarianPizza

Asserted

- owl:Thing
 - :Pizza
 - :CheesyPizza
 - :FourCheesePizza
 - :HighCaloriePizza
 - :InterestingPizza
 - :LowCaloriePizza
 - :NamedPizza
 - :NonVegetarianPizza**
 - :SpicyPizza
 - :VegetarianPizza
 - :PizzaBase
 - :PizzaTopping
 - :ValuePartition

:NonVegetarianPizza — http://www.pizza.com/ontologies/

Annotations Usage

Annotations: :NonVegetarianPizza

Annotations +

Description: :NonVegetarianPizza

Equivalent To +

SubClass Of +

- :Pizza

General class axioms +

SubClass Of (Anonymous Ancestor)

- hasBase some :PizzaBase
- hasCalorificContentValue some xsd:integer

Instances +

Target for Key +

Disjoint With +

- :VegetarianPizza

Exercise 53

Classes

Object properties

Data properties

Annotations

Usage

Class hierarchy: :NonVegetarianPizza

Annotations: :NonVegetarianPizza

Annotations +

owl:Thing

:Pizza

:CheesyPizza

:FourCheesePizza

:HighCaloriePizza

:InterestingPizza

:LowCaloriePizza

:NamedPizza

:NonVegetarianPizza

:SpicyPizza

:VegetarianPizza

:PizzaBase

:PizzaTopping

:ValuePartition

Description: :NonVegetarianPizza

Equivalent To +

SubClass Of +

:Pizza

and (not (:VegetarianPizza))

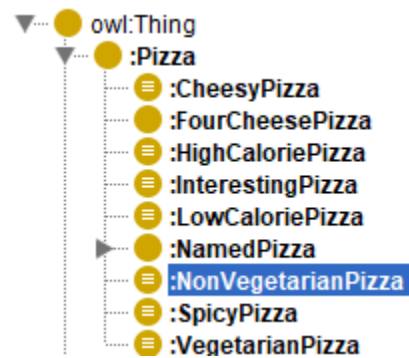
General class axioms +

SubClass Of (Anonymous Ancestor)

hasBase some :PizzaBase

hasCalorificContentValue some xsd:integer

Exercise 54



Exercise 55

Classes

Object properties

Data properties

Class hierarchy: :AmericanHotPizza

owl:Thing

:Pizza

:CheesyPizza

:FourCheesePizza

:HighCaloriePizza

:InterestingPizza

:LowCaloriePizza

:NamedPizza

:AmericanaPizza

:AmericanHotPizza

:MargheritaPizza

:SohoPizza

:NonVegetarianPizza

:SpicyPizza

:VegetarianPizza

:PizzaBase

:PizzaTopping

:ValuePartition

Annotations

Usage

Annotations: :AmericanHotPizza

Annotations +

Description: :AmericanHotPizza

Equivalent To +

SubClass Of +

:NamedPizza

hasTopping only

(:JalapenoPepperTopping or :MozzarellaTopping or :PepperoniTopping or :TomatoTopping)

hasTopping some :JalapenoPepperTopping

hasTopping some :MozzarellaTopping

hasTopping some :PepperoniTopping

hasTopping some :TomatoTopping

:CheesyPizza

:InterestingPizza

:NonVegetarianPizza

:SpicyPizza

Exercise 56



Exercise 57

Classes

Object properties

Data properties

Class hierarchy: :UnclosedPizza

owl:Thing

:Pizza

:CheesyPizza

:FourCheesePizza

:HighCaloriePizza

:InterestingPizza

:LowCaloriePizza

:NamedPizza

:AmericanaPizza

:AmericanHotPizza

:MargheritaPizza

:SohoPizza

:UnclosedPizza

:NonVegetarianPizza

:SpicyPizza

:VegetarianPizza

:PizzaBase

:PizzaTopping

:ValuePartition

Annotations

Usage

Annotations: :UnclosedPizza

Annotations +

Description: :UnclosedPizza

SubClass Of +

:NamedPizza

hasTopping some :MozzarellaTopping

:CheesyPizza

General class axioms +

SubClass Of (Anonymous Ancestor)

hasBase some :PizzaBase

hasCalorificContentValue some xsd:integer

:Pizza

and (hasTopping some :CheeseTopping)

Exercise 58

Individuals: :QuattroFormaggio

+

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:America

:England

:ExampleAmericana

:ExampleMargherita

:http://www.pizza.com/ontologies/pizza.owl#ExampleAmericana

:France

:Germany

:Italy

:QuattroFormaggio

Classes

Object properties

Data properties

Class hierarchy: :Country

owl:Thing

:Country

:Pizza

:PizzaBase

:PizzaTopping

:ValuePartition

Asserted

Annotations

Usage

Annotations: :Country

Annotations +

Description: :Country

Equivalent To +

SubClass Of +

owl:Thing

General class axioms +

SubClass Of (Anonymous Ancestor)

Instances +

:America

:England

:France

:Germany

:Italy

Exercise 59

Classes	Object properties	Data properties	Annotations	Usage
Class hierarchy: :MozzarellaTopping			Annotations: :MozzarellaTopping	
owl:Thing :Country :Pizza :PizzaBase :PizzaTopping :CheeseTopping :MozzarellaTopping :ParmesanTopping :ProbelnconsistentTopping :MeatTopping :SeafoodTopping :VegetableTopping :ValuePartition			Annotations + Description: :MozzarellaTopping Equivalent To + SubClass Of + :CheeseTopping hasCountryOfOrigin value :Italy hasSpiciness some :Mild	

Exercise 60

Classes	Object properties	Data properties	Annotations	Usage
Class hierarchy: :Country			Annotations: :Country	
owl:Thing :Country :Pizza :PizzaBase :PizzaTopping :ValuePartition			Annotations + Description: :Country Equivalent To + SubClass Of + {:America , :England , :France , :Germany , :Italy} owl:Thing	